

Composite Area and Perimeter Challenges

Answer Key

Grade 3

Quick Answers

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|----------------------|---|
| 1. 24 cm^2 | 8. 28 cm |
| 2. 24 m | 9. missing top side = 5, missing short side = 2,
(b) perimeter = 28 cm |
| 3. 8 square units | 10. 26 m^2 |
| 4. 28 cm | 11. 24 m |
| 5. 30 m^2 | 12. (a) area = 36 m^2 , (b) perimeter = 32 m, (c)
= > |
| 6. 20 m^2 | |
| 7. 45 cm^2 | |
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Curriculum

Level: Grade 3

3.MD.C.7 – problems 1, 3, 5, 6, 7, 10, 12

3.MD.D.8 – problems 2, 4, 8, 9, 11, 12

Difficulty 1–5 of 5 across 16 tagged checks.

1. A rectangle is 6 cm long and 4 cm wide. What is its area?

Solution: Area is length times width: $6 \times 4 = 24$. The area is

24 cm^2

2. A pen is 9 m long and 3 m wide. What is its perimeter?

Solution: Perimeter adds all four sides: $2 \times (9 + 3) = 24$. The perimeter is

24 m

3. Each grid square is one square unit. Find the area of the L-shape.

Solution: Split along the grid: the bottom bar is 4 squares long and 1 square tall (4 squares), and the left column above it is 2 squares wide and 2 squares tall (4 squares). Together: $4 + 4 = 8$, so the area is

8 square units

4. Walk around the L-shaped patio with sides 8, 3, 4, 3, 4, and 6 cm. What is the perimeter?

Solution: Add every side as you walk around the edge: $8 + 3 + 4 + 3 + 4 + 6 = 28$. The perimeter is

28 cm

5. Find the area of the L-shaped playground (sides 10, 2, 5, 2, 5, 4 m).

Solution: Split along the dashed line: the bottom rectangle is 10 m by 2 m ($10 \times 2 = 20$) and the top-left rectangle is 5 m by 2 m ($5 \times 2 = 10$). Total area: $20 + 10 = 30$, so the playground covers

30 m^2

6. An L-shaped rug joins a 7 m by 2 m piece and a 3 m by 2 m piece. What is the total area?

Solution: Find each rectangle's area, then add: $7 \times 2 = 14$ and $3 \times 2 = 6$, so $14 + 6 = 20$.
The rug covers 20 m^2

7. Find the area of the L-shaped sticker (sides 10, 3, 5, 3, 5, 6 cm).

Solution: Split along the dashed line: the bottom rectangle is 10 cm by 3 cm ($10 \times 3 = 30$) and the top-left rectangle is 5 cm by 3 cm ($5 \times 3 = 15$). Total: $30 + 15 = 45$, so the area is 45 cm^2

8. A photo frame is 8 cm by 6 cm. How many centimeters of wooden edge go around it?

Solution: "Around" means perimeter: $2 \times (8 + 6) = 28$. The edge is 28 cm long.

9. Missing sides! The shape shows sides 9, 3, 4, ?, ?, 5 cm. (a) Find the two missing sides. (b) Find the perimeter.

Solution (a): The two top pieces together must match the 9 cm bottom, so the missing top side is $9 - 4 = 5$ cm. 5

The two right-hand pieces together must match the 5 cm left side, so the missing short side is $5 - 3 = 2$ cm. 2

Solution (b): Now add every side around the shape: $9 + 3 + 4 + 2 + 5 + 5 = 28$. The perimeter is 28 cm

10. A deck joins a 6 m by 3 m rectangle and a 4 m by 2 m rectangle. What is the total area?

Solution: Add the two rectangle areas: $6 \times 3 = 18$ and $4 \times 2 = 8$, so $18 + 8 = 26$. The deck covers 26 m^2

11. The staircase shape has a 6 m bottom, a 6 m left side, and every step piece is 2 m. What is the perimeter?

Solution: Walk around: the bottom is 6, then six step pieces of 2 each ($2 + 2 + 2 + 2 + 2 + 2 = 12$), then the left side of 6. Add them: $6 + 12 + 6 = 24$. The perimeter is 24 m

12. Challenge. The garden has sides 12, 2, 6, 2, 6, 4 m. (a) Area? (b) Perimeter? (c) Compare the two numbers.

Solution (a): Split along the dashed line: bottom rectangle 12 m by 2 m ($12 \times 2 = 24$) plus top-left rectangle 6 m by 2 m ($6 \times 2 = 12$): $24 + 12 = 36$. The area is 36 m^2

Solution (b): Add every side: $12 + 2 + 6 + 2 + 6 + 4 = 32$. The perimeter is 32 m

Solution (c): 36 is more than 32, so $36 > 32$

— here the area number is bigger, but that is not true for every shape!